

Digital Signal Processing - Lecture 5

Spectrograms, the short-time Fourier transform, and windowing

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| Overview

This lecture

01 Why spectrograms?

02 How the STFT works

03 Windowing

04 Exercises

| Why spectrograms?

Listen to the two sounds

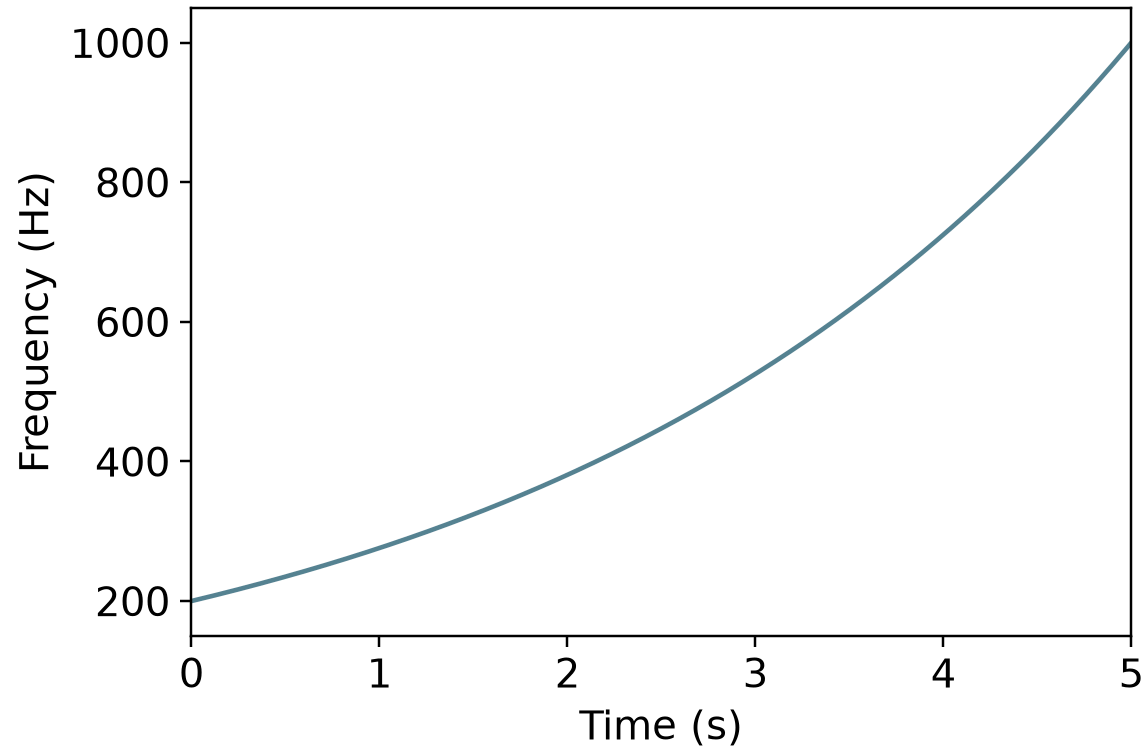
Sound A

Sound B

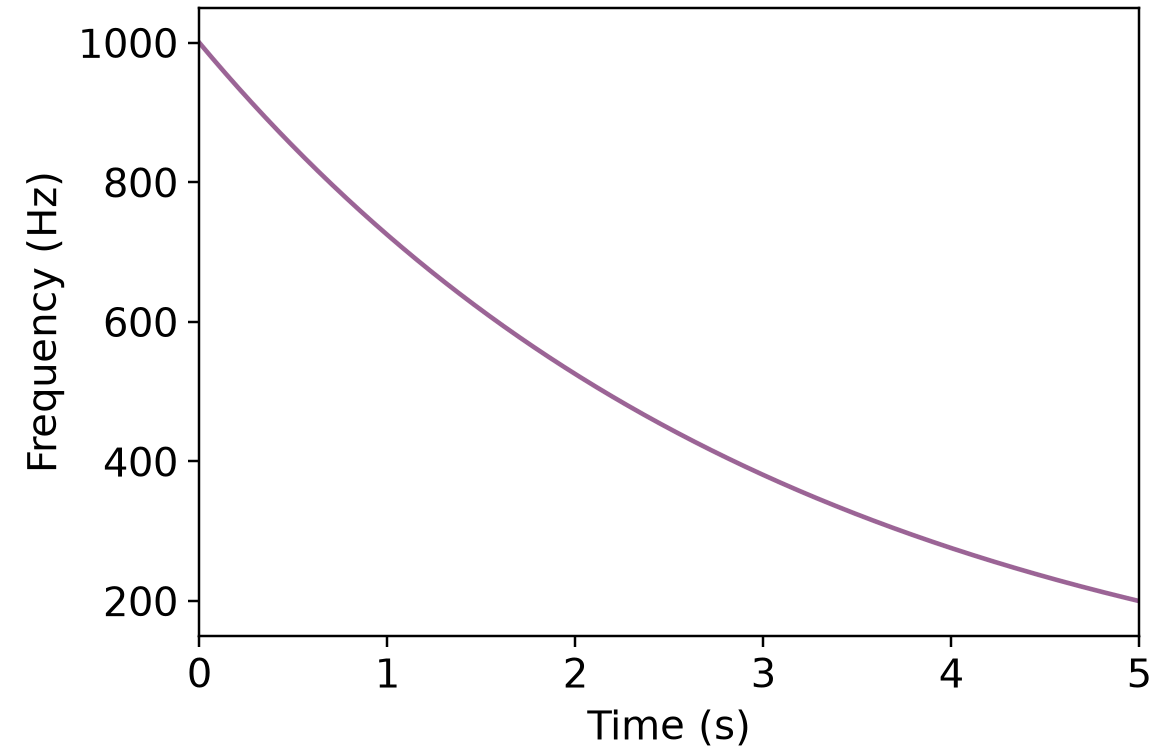
How does the pitch change in each sound?

Which graph is which sound?

Frequency vs time

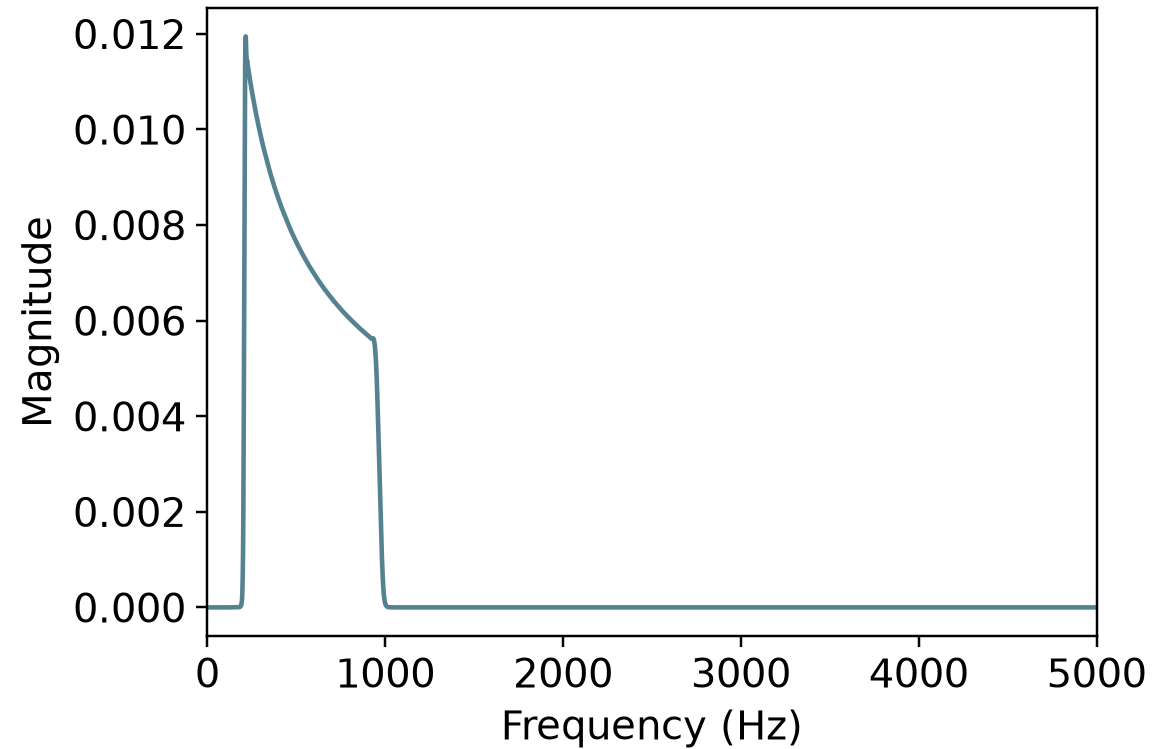


Frequency vs time

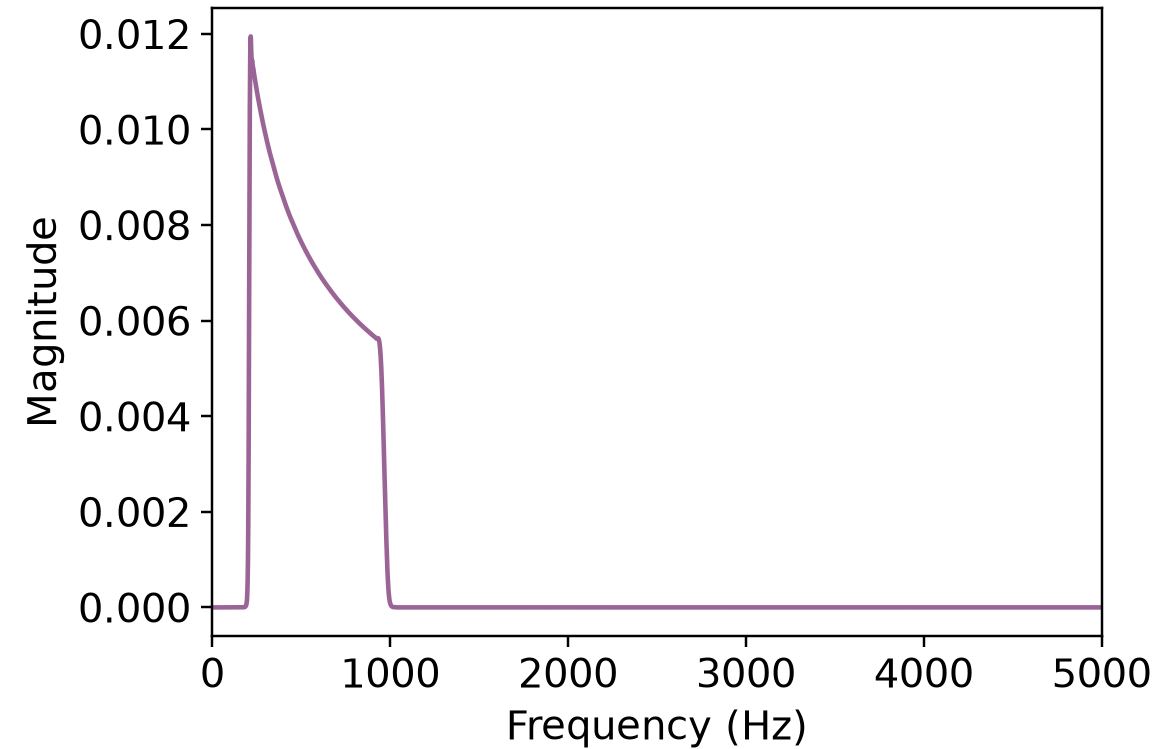


Can the spectrum distinguish them?

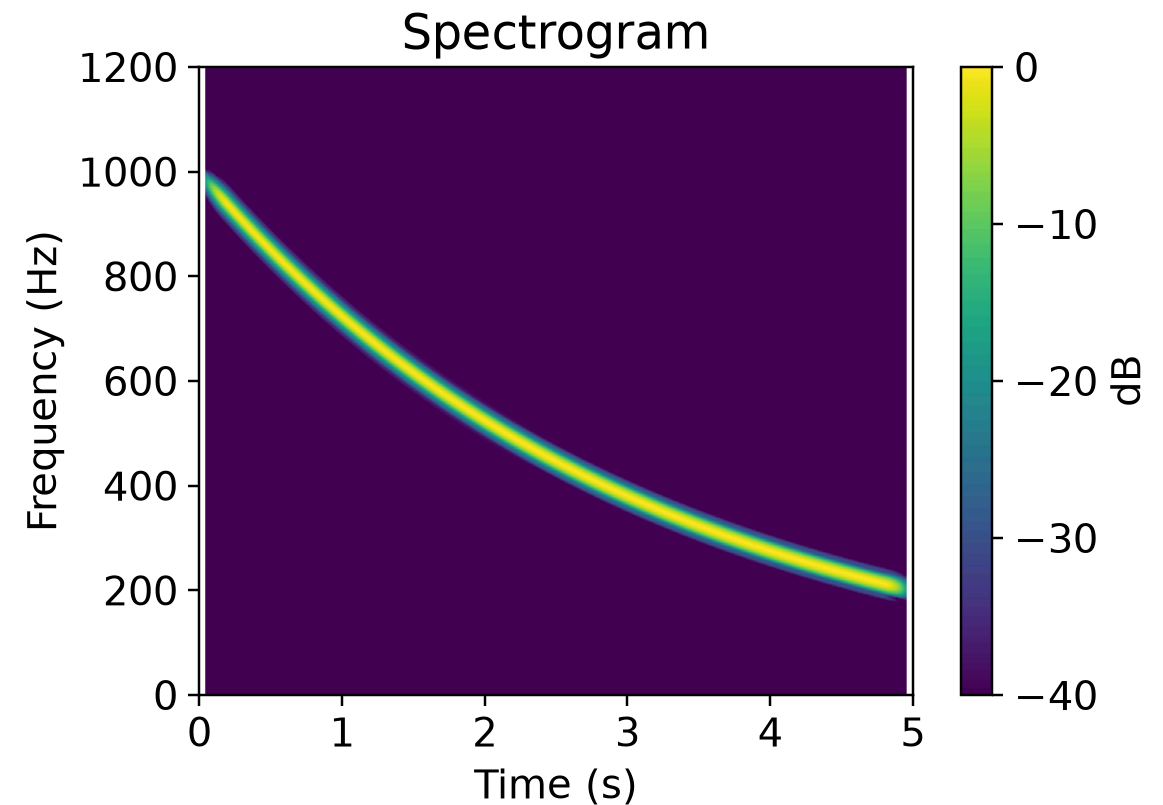
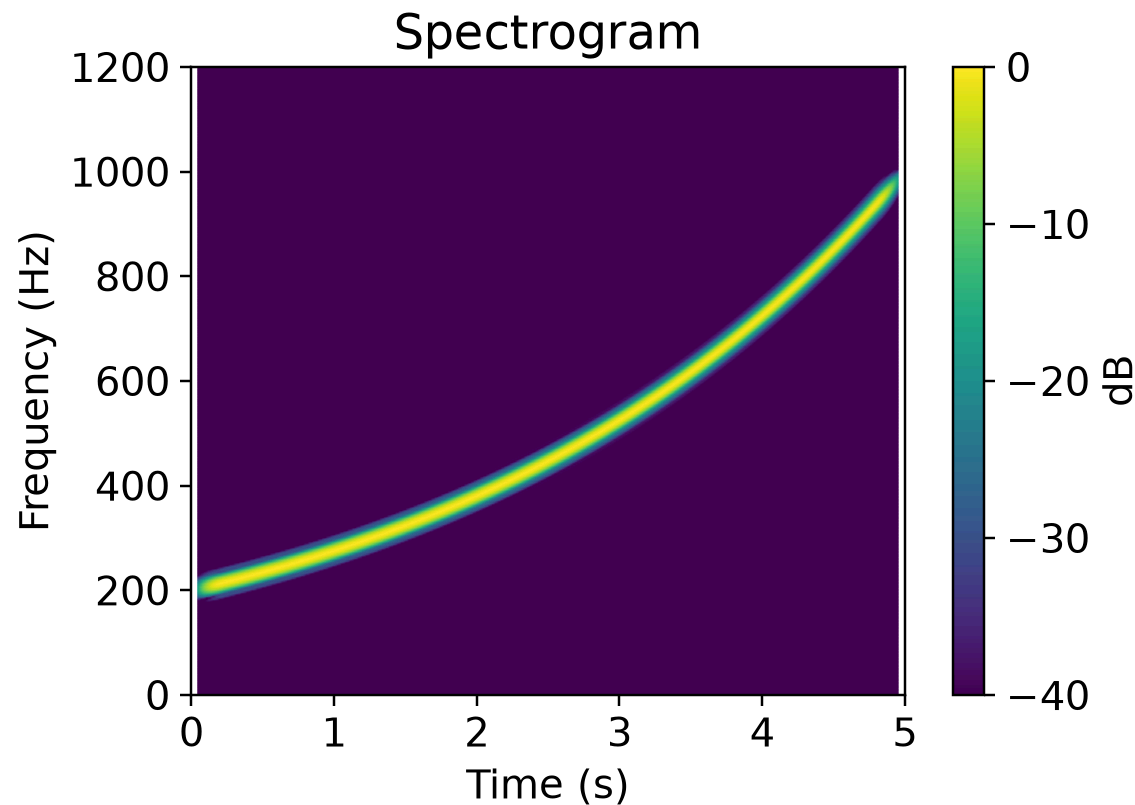
Spectrum plot



Spectrum plot



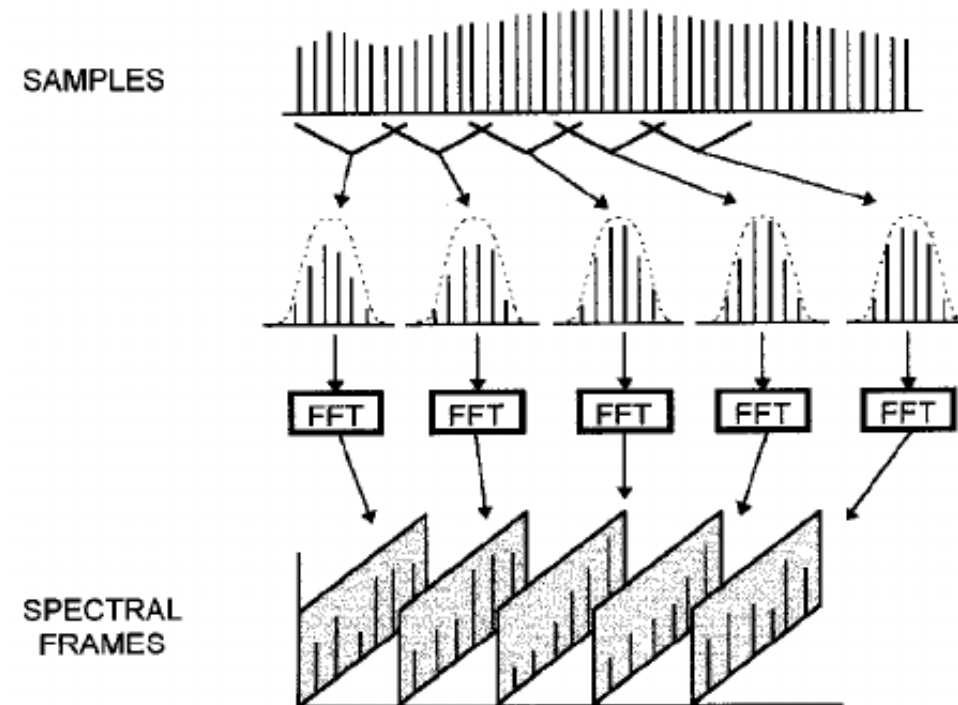
Frequency content over time



Time in horizontal axis, frequency in vertical axis. Brighter colours show stronger frequency components.

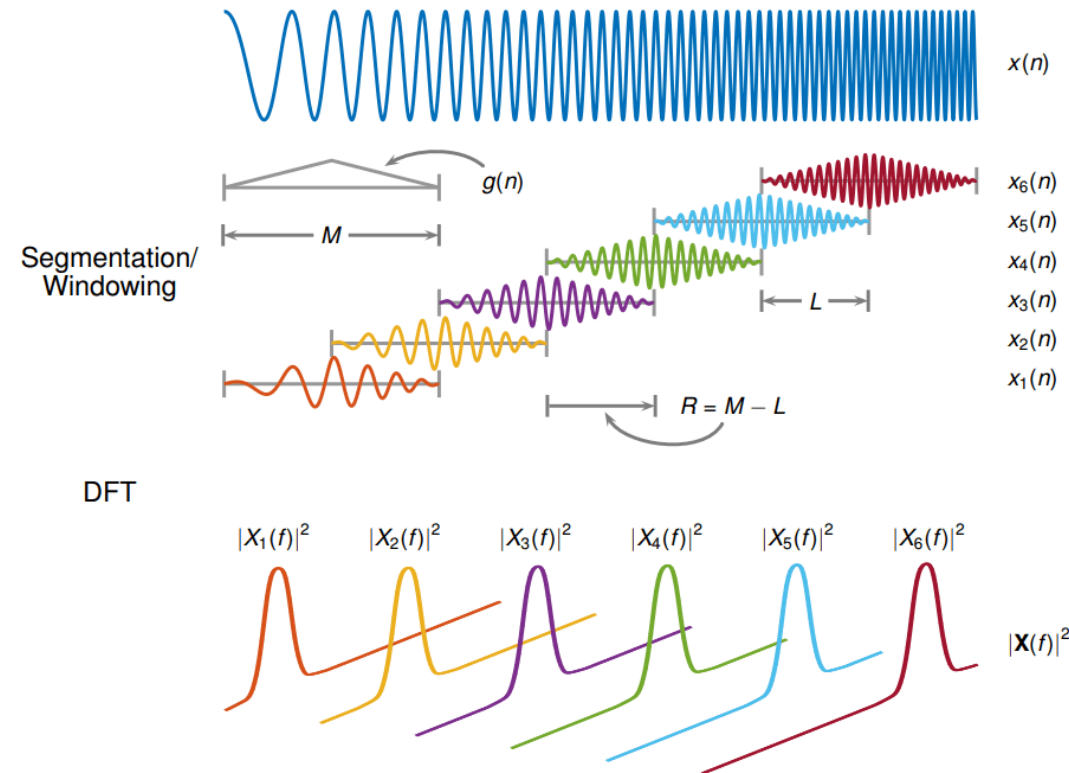
| Short-time Fourier transform

Short-time Fourier transform



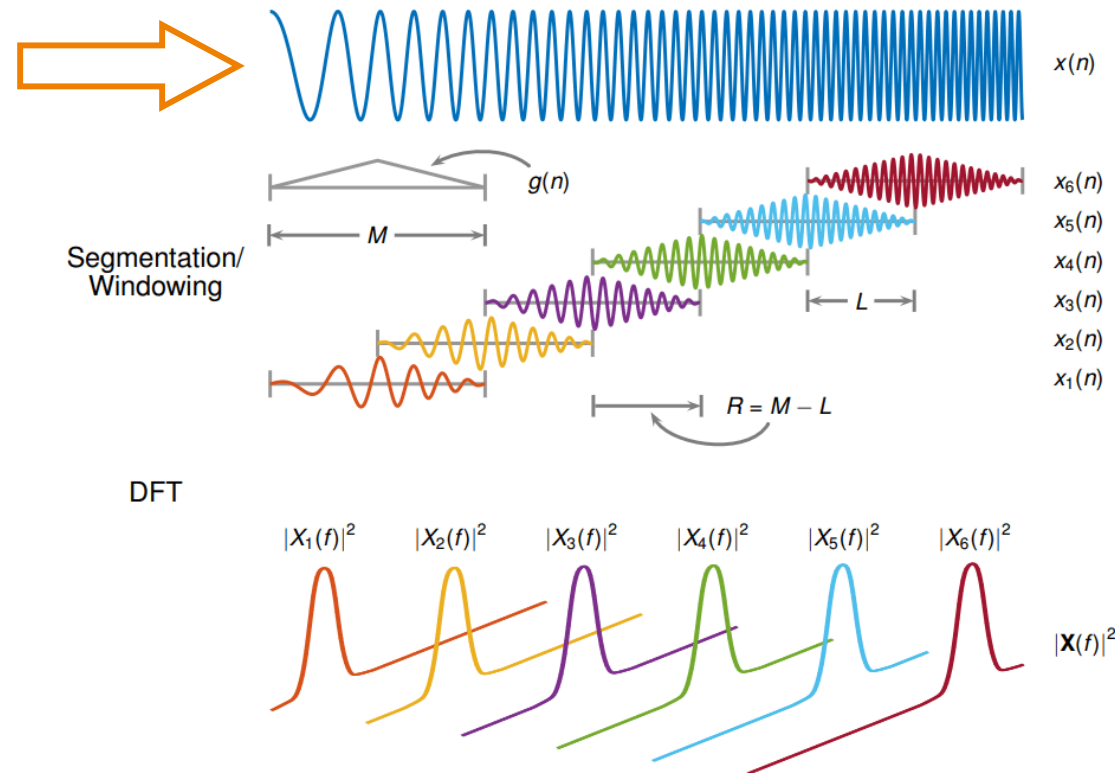
Source: https://www.researchgate.net/figure/Short-time-Fourier-transform-STFT-with-envelope-and-two-sample-overlap_fig7_231828310

Short-time Fourier transform



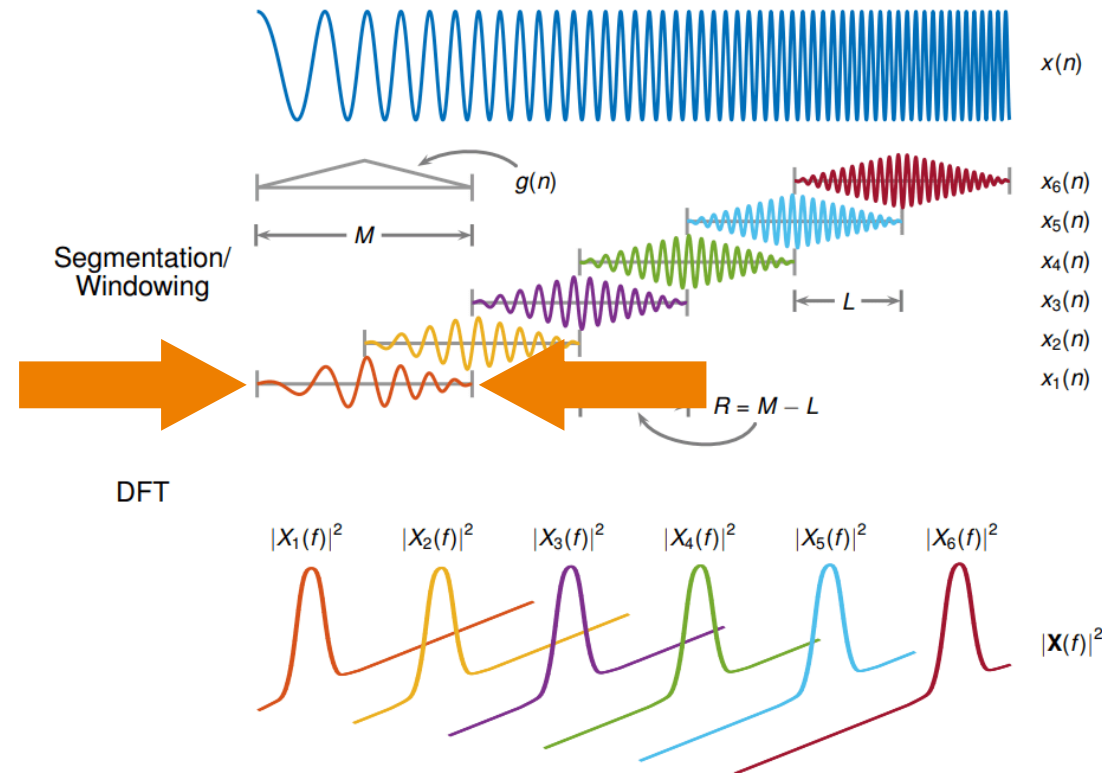
Source: <https://nl.mathworks.com/help/signal/ref/stft.html>

The input signal



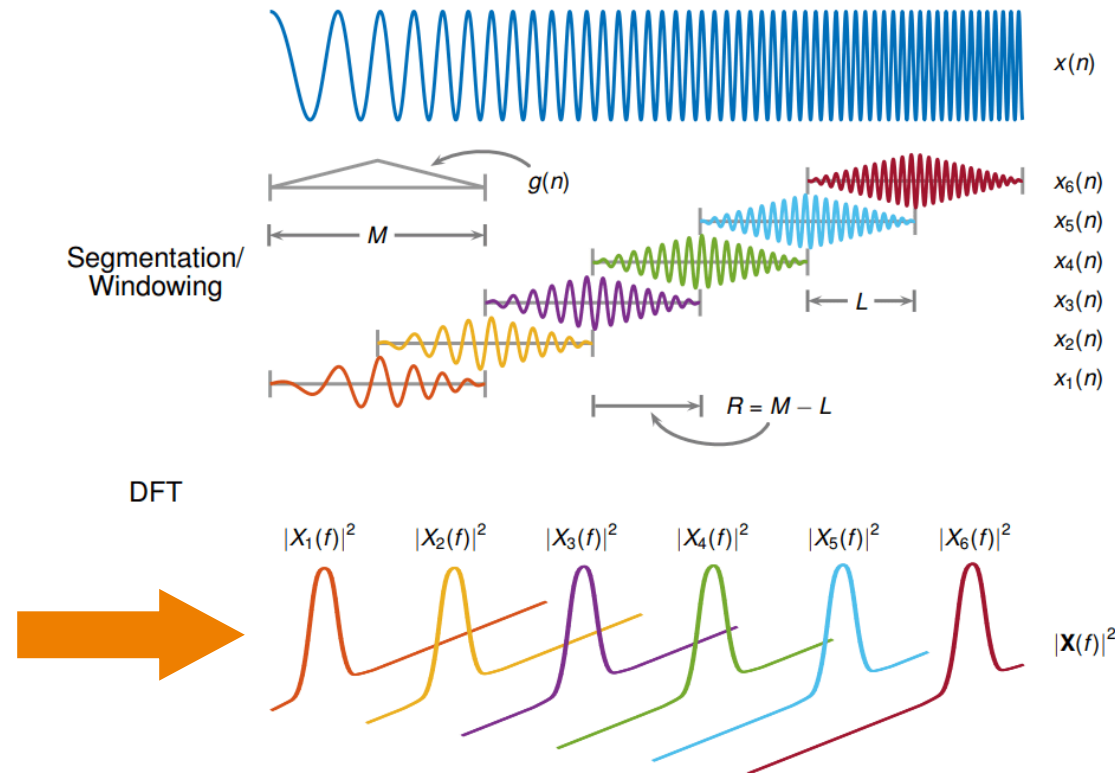
Source: <https://nl.mathworks.com/help/signal/ref/stft.html>

Select a short segment



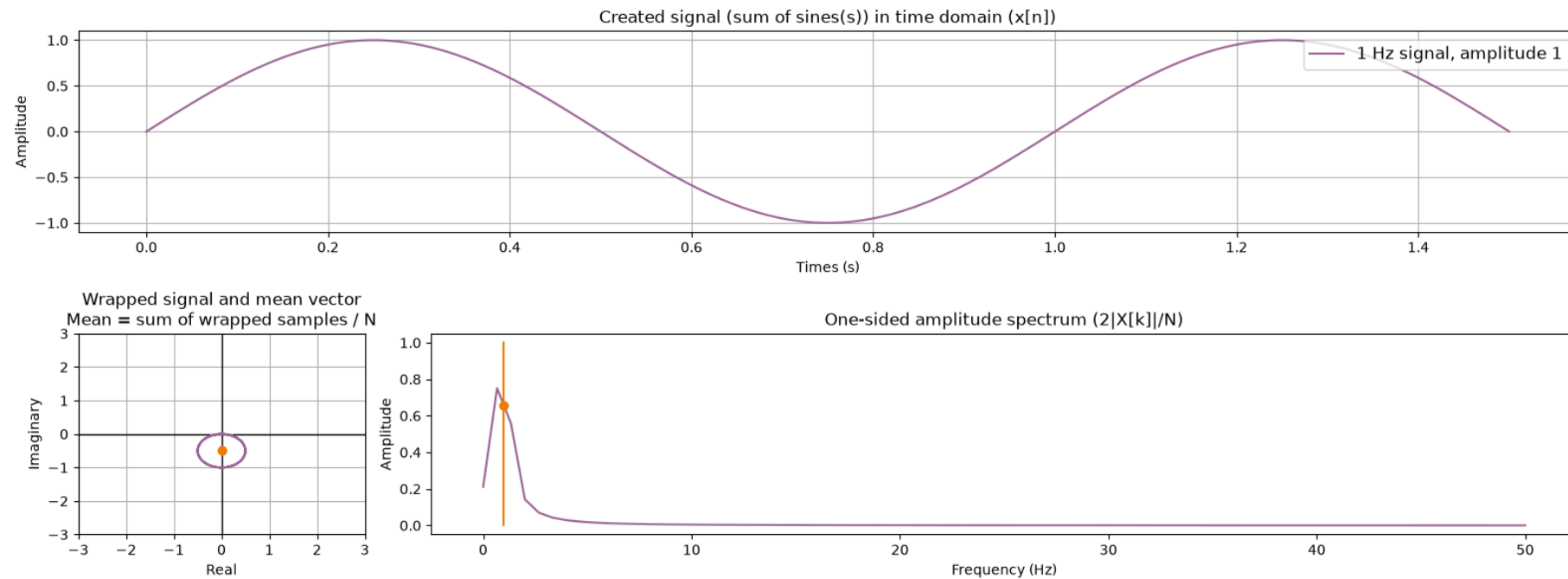
Source: <https://nl.mathworks.com/help/signal/ref/stft.html>

Fourier Transform the segment. But...

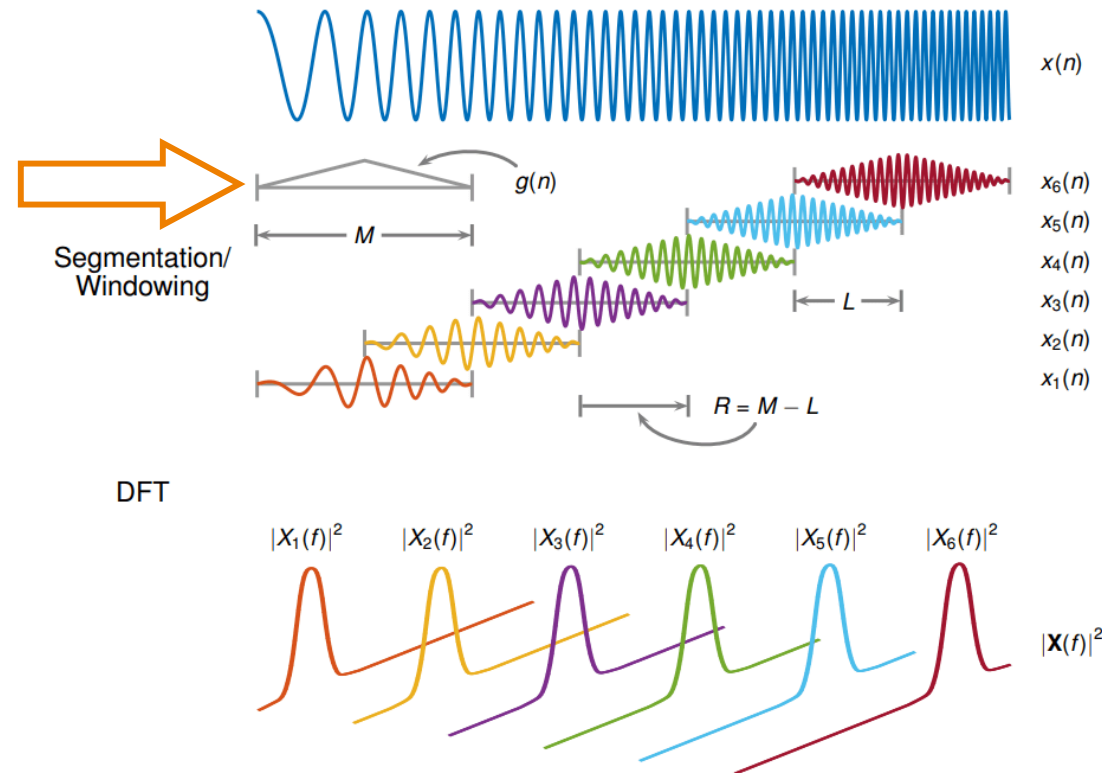


Source: <https://nl.mathworks.com/help/signal/ref/stft.html>

Recall spectral leakage

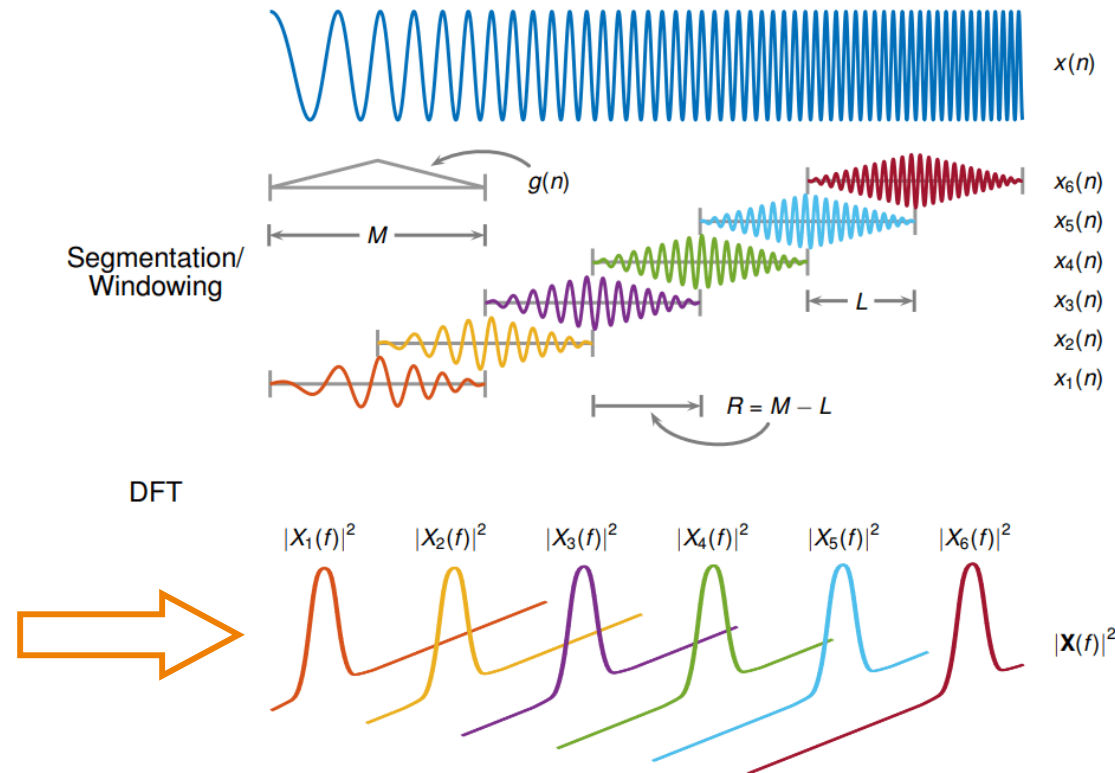


Apply the window



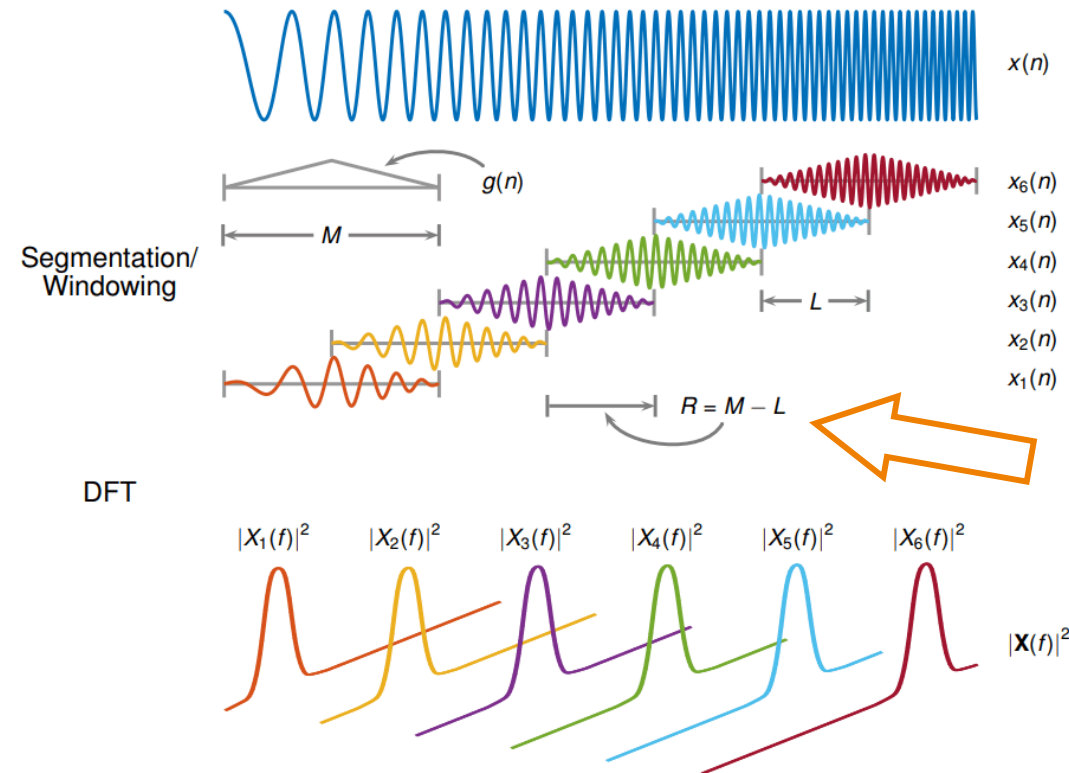
Source: <https://nl.mathworks.com/help/signal/ref/stft.html>

Transform each segment



Source: <https://nl.mathworks.com/help/signal/ref/stft.html>

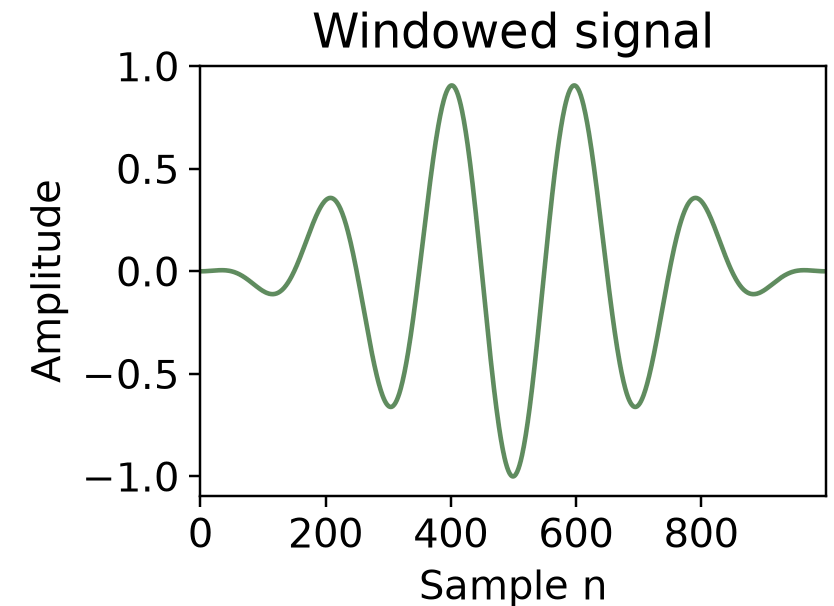
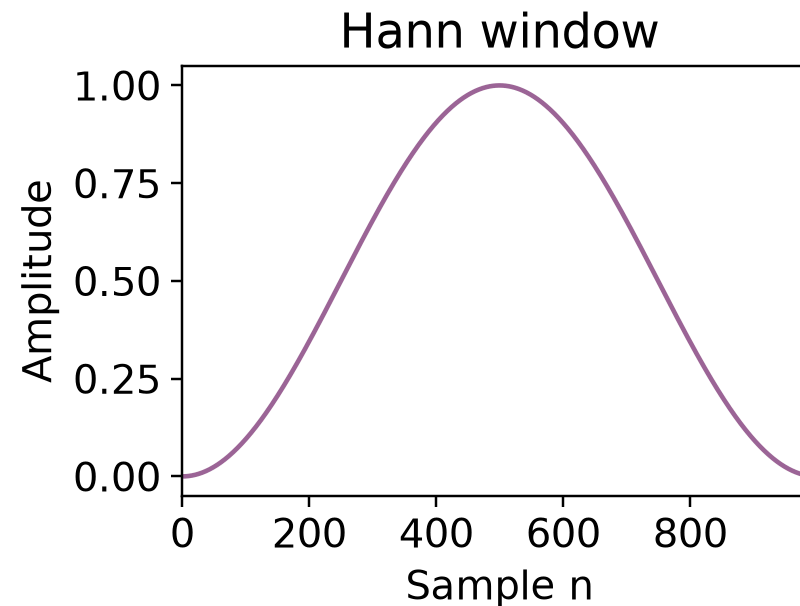
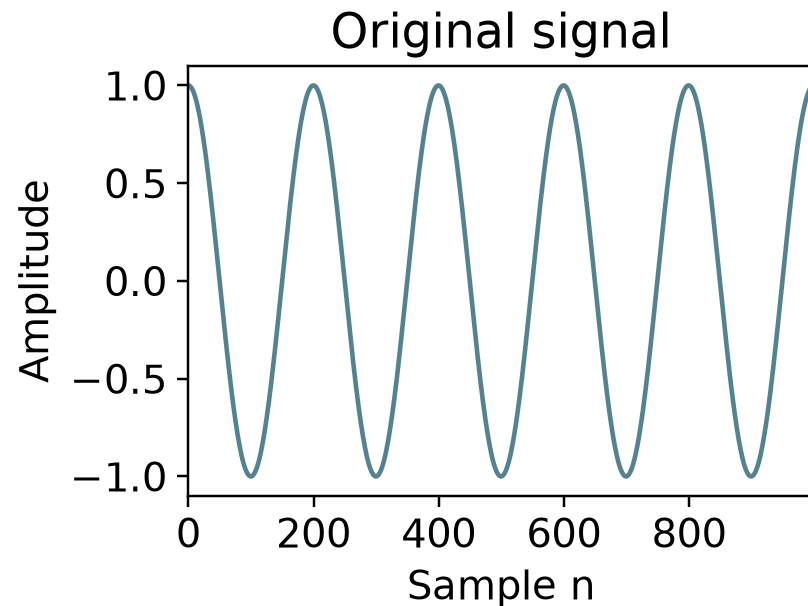
Note the overlap



Source: <https://nl.mathworks.com/help/signal/ref/stft.html>

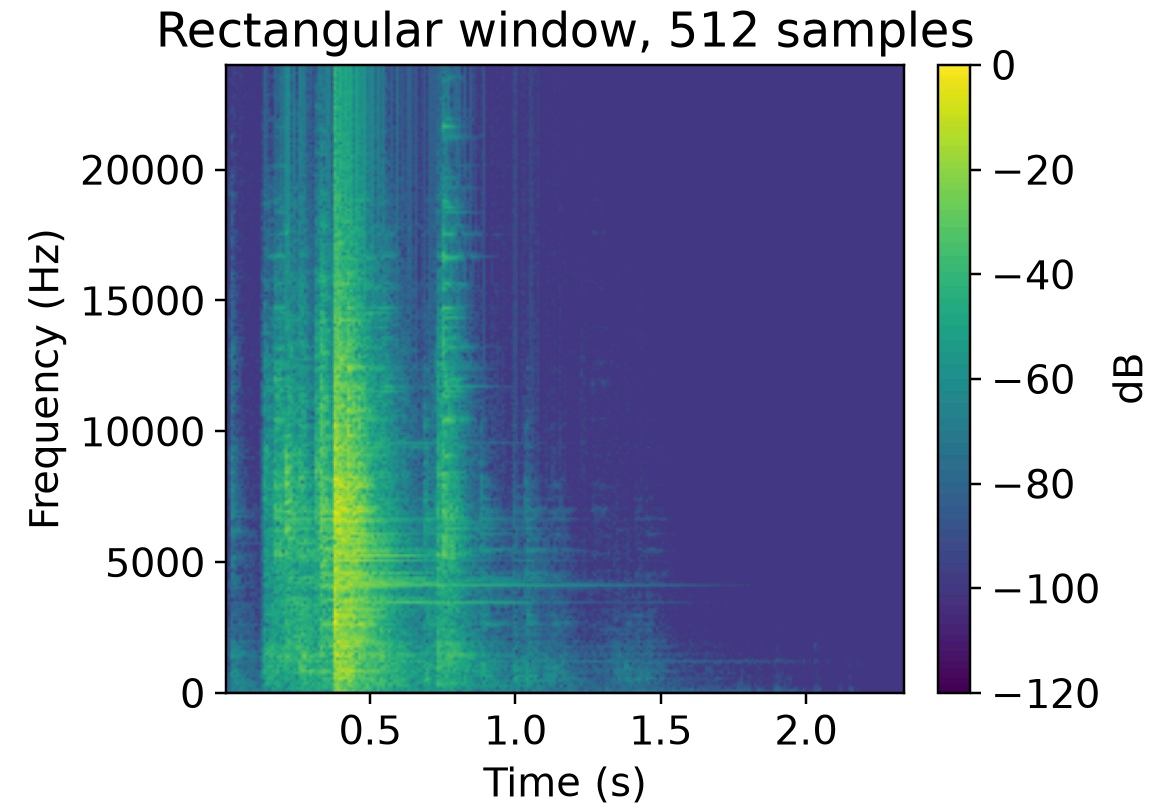
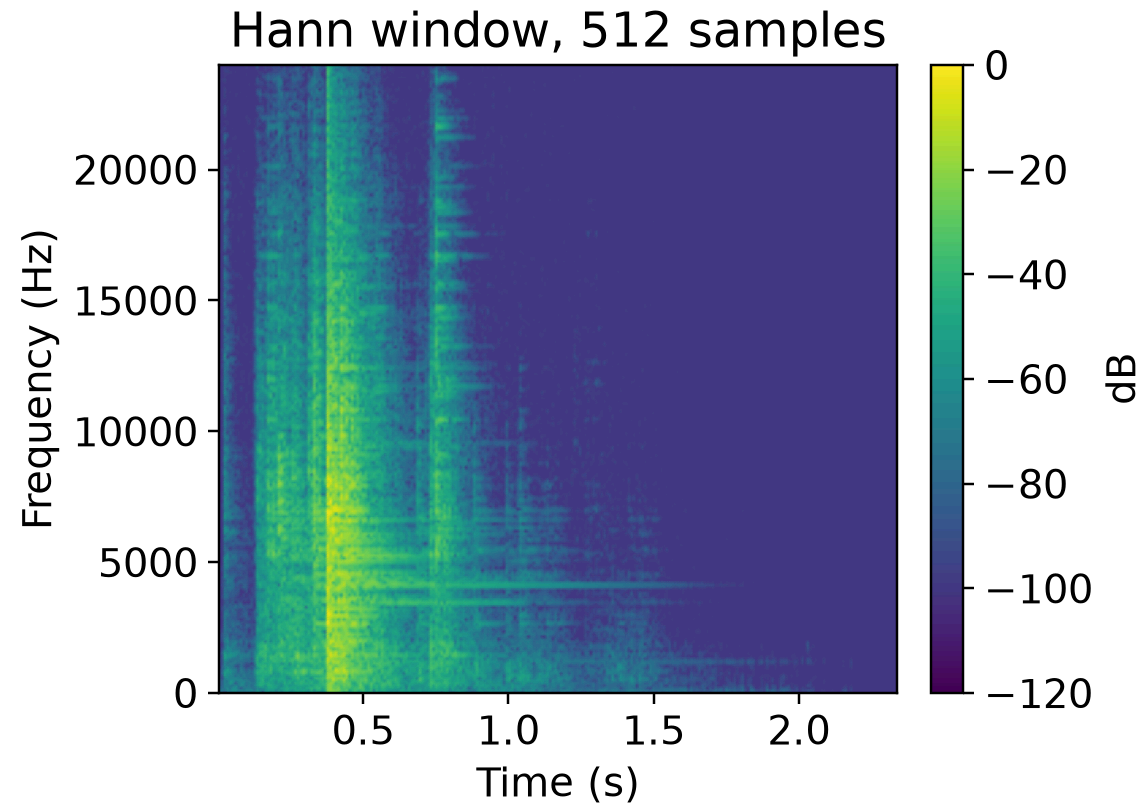
| Windowing

Windowing the signal



Multiply the signal segment by the window before taking its DFT.

Hann and rectangular windows



A Hann window reduces spectral leakage by smoothing segment edges, at the cost of broader frequency peaks.

| Exercises

Exercises!

`../exercises/DSP-Lecture-05-Exercises.ipynb`